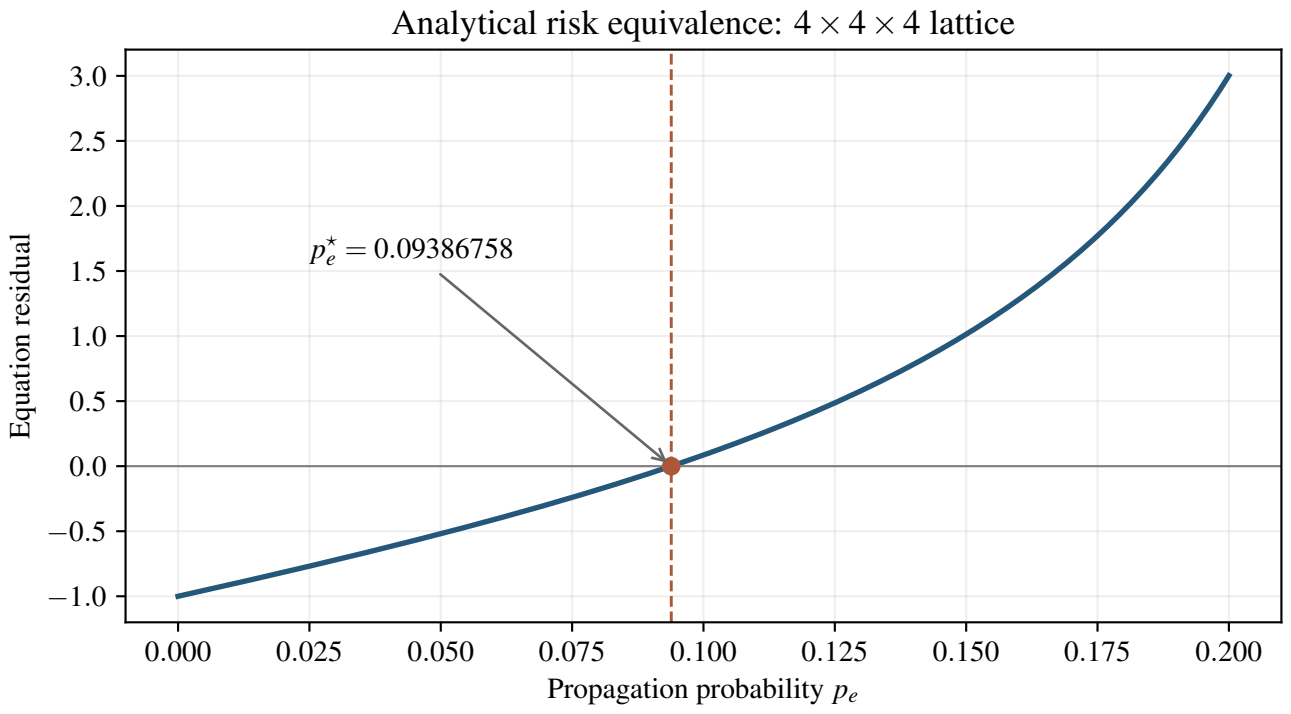
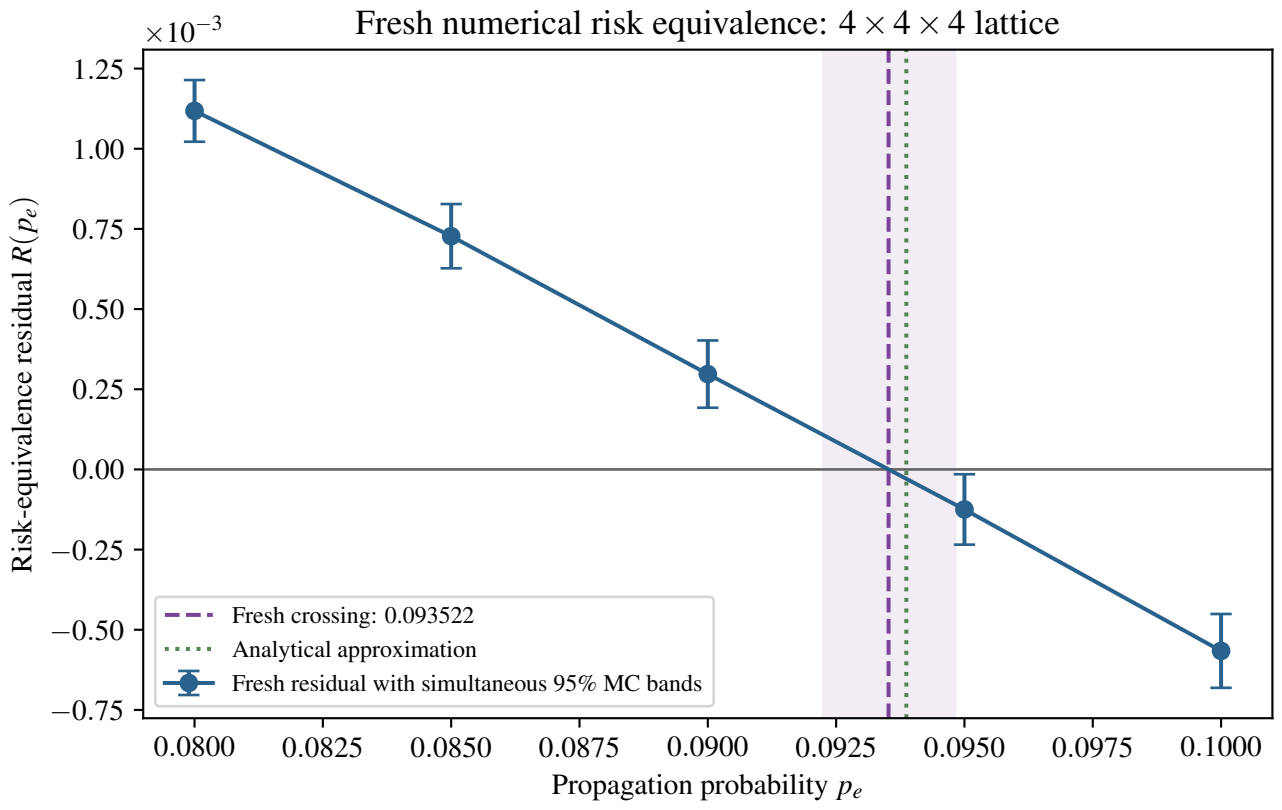
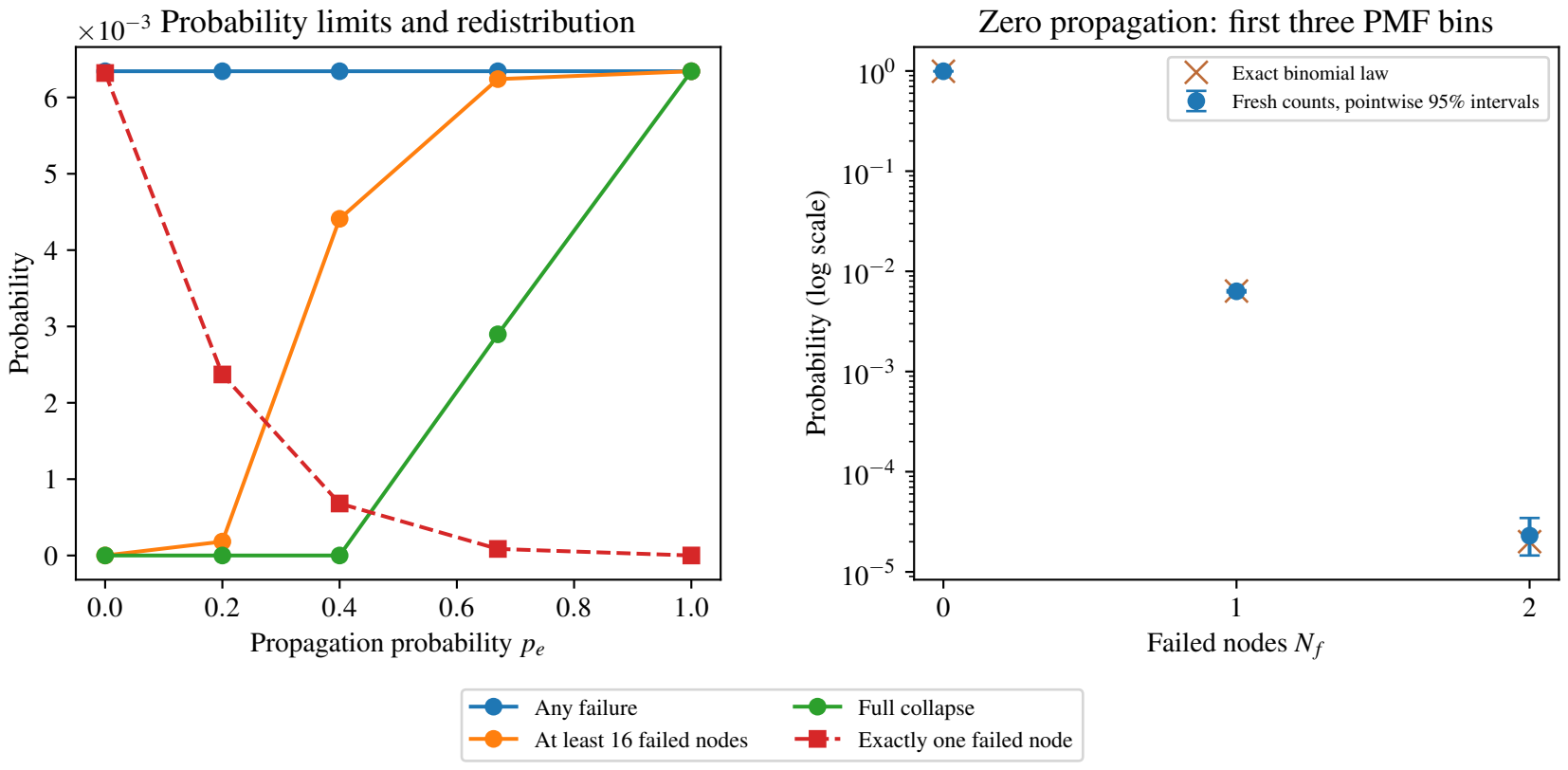

Analytical risk equivalence: $4 \times 4 \times 4$ lattice
$p_e^\star = 0.09386758$
Equation residual
Propagation probability p_e
3.0
2.5
2.0
1.5
1.0
0.5
0.0
−0.5
−1.0
0.000
0.025
0.050
0.075
0.100
0.125
0.150
0.175
0.200


Fresh numerical risk equivalence: $4 \times 4 \times 4$ lattice
$\times 10^{-3}$
1.25
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
Risk-equivalence residual $R(p_e)$
Fresh crossing: 0.093522
Analytical approximation
Fresh residual with simultaneous 95% MC bands
0.0800
0.0825
0.0850
0.0875
0.0900
0.0925
0.0950
0.0975
0.1000
Propagation probability p_e


Probability limits and redistribution
$\times 10^{-3}$
Probability
Propagation probability p_e
0.0
0.2
0.4
0.6
0.8
1.0
0
1
2
3
4
5
6
Zero propagation: first three PMF bins
Probability (log scale)
Failed nodes N_f
Exact binomial law
Fresh counts, pointwise 95% intervals
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
Any failure
At least 16 failed nodes
Full collapse
Exactly one failed node